SUZHE XU

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EDUCATION

Peking University Jun. 2025 – Present
Visiting Student / Research Assistant, Wangxuan Institute of Computer Technology
Huaqiao University Sep. 2023 – Jun. 2027 (expected)
Undergraduate, Artificial Intelligence, School of Computer Science and Technology
Rank: Comprehensive rank (2023–2025): **1/40**; GPA rank: **4/40**.
GPA: 4.09/5.00 (84.77/100).
National Scholarship, Ministry of Education of China, 2024 — awarded to approximately **0.3%** of university students nationwide.

RESEARCH INTERESTS

Efficient generative modeling: native few-step generation, diffusion distillation, video inverse problems, and streaming world models; training stability and multimodal post-training.

PUBLICATIONS AND MANUSCRIPTS

1. Weimin Bai*, **Suzhe Xu***, Yiwei Ren, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR: Real-Time Video Inverse Problem Solver with Distilled Diffusion Prior.** *CVPR*, 2026. **Highlight**. Co-project lead and presenter. *Equal contribution. [Paper] [Project] [Presentation]
2. **Suzhe Xu***, Weimin Bai*, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR++: Improved Posterior Distillation for Real-Time Video Inverse Problem Solving.** *Submitted to IEEE Transactions on Multimedia*, 2026. *Equal contribution.
3. **KAT-Coder-V2 Technical Report.** arXiv:2603.27703, 2026. [Report]
   Contributions: execution-based WebCoding data curation, supervised fine-tuning, and long-context continued pretraining.

RESEARCH EXPERIENCE

Peking University Jun. 2025 – Present
Advisors: Wenzheng Chen and He Sun

InstantViR and InstantViR++
Real-time video inverse problems

- To enable real-time video restoration without iterative posterior sampling, co-led InstantViR, distilling a video diffusion prior into a causal one-step solver for deblurring, inpainting, and super-resolution.
- The published system exceeds **35 FPS**, with up to **100× speedup** over iterative solvers, without external paired clean/degraded training videos.
- Investigated how teacher–student latent-space mismatch and competing objectives affect posterior distillation in InstantViR++. Used bidirectional latent conversion and cyclic loss weighting to balance measurement consistency and generative supervision, and evaluated the design through controlled comparisons. Manuscript submitted to **IEEE Transactions on Multimedia**.

RESEARCH EXPERIENCE (CONTINUED)

Streaming World Models for Autonomous Driving Jun. 2026 – Present
Industry-collaborative research project Peking University

- Investigated few-step causal generation of driving videos through chunk-causal fine-tuning, **four-step causal consistency distillation**, and Self-Forcing DMD in a three-view pipeline.
- Used self-generated-history rollouts and clean-history KV caching to study train–test mismatch; compared CD and 25-/49-frame SF-DMD for temporal variation, pixel fidelity, and later-chunk errors.
- Also implemented single-view generation; **five-view, 101-frame** causal distillation is ongoing.

Prometheus: Causal Few-Step Streaming World Models for Autonomous Driving. Working title. Manuscript in preparation; targeting submission to CVPR 2027.

Multi-Agent World Model Post-Training and Evaluation Ongoing
Research project Peking University

- To model multiple agents with independent action inputs, adapted Cosmos3-Edge for native dual-view generation using agent-specific action tokens and four-axis positional encoding. Trained with a **93-to-189-frame curriculum** to extend the prediction horizon.
- Aligned frame rates, prediction horizons, and sampling budgets for controlled comparisons with the Gamma teacher. Found persistent material and geometric errors despite lower validation loss, revealing a gap between the training metric and visual fidelity.

Research in progress; targeting submission to CVPR 2027.

Rednote (Xiaohongshu), Dots Studio Aug. 2026 – Oct. 2026
Multimodal Research Intern — Ace Top Talent Internship Program Mentor: Weijian Luo

- **Native few-step language generation:** Investigate how few-step generative objectives transfer to language modeling through FLM cross-entropy/MSE baselines, MeanFlow, and simplified consistency training. Compare **1/2/4-step generation** using Gen PPL, MAUVE, repetition rates, and samples to examine objective–quality discrepancies and quality–diversity–compute trade-offs.
- **Distribution-matching distillation:** Investigate how score-difference processing affects one-step diffusion distillation using norm-preserving spectral transformations and rotation-based adaptations. Use controlled CIFAR-10 comparisons to disentangle residual-space processing, auxiliary score estimation, and student optimization, assessing sample quality, diversity, and convergence.
- Coordinate controlled studies across parallel **8-GPU experiments**(**100+ GPUs** in aggregate at peak).

ADDITIONAL INDUSTRY EXPERIENCE

Kuaishou Jan. 2026 – Jun. 2026
Algorithm Research Intern, Frontier Agent Algorithms Mentors: Han Zhang and Jinhua Hao

- To improve the functional reliability of WebCoding training data, combined LLM-generated tests with browser-based execution and UI-agent verification. Curated **10k+ web-tool/game examples**, participated in supervised fine-tuning, and extended evaluation to 3D scenarios.
- To support repository-level editing and long-context code modeling, derived fill-in-the-middle and unified-diff training examples from issue–pull request pairs, yielding **187,858 examples from 3,600+ repositories**. Conducted continued pretraining and evaluation of Qwen3.5-35B-A3B at **64K context**.

*Also conducted InstantViR++ research with Jinhua Hao (**Kling AI**) and Peking University during this internship, supported by Kuaishou; see research experience above.*

SKILLS AND LANGUAGES

Technical Skills: Python; PyTorch; distributed training (FSDP, DeepSpeed, Megatron-Bridge); LoRA; TensorBoard; Git; LaTeX.
Languages: Mandarin Chinese; English (IELTS 7.5 overall); French (B1, self-assessed).